

PKA regulation of neuronal function requires the dissociation of catalytic subunits from regulatory subunits

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Abstract

Protein kinase A (PKA) plays essential roles in diverse cellular functions. However, the spatiotemporal dynamics of endogenous PKA upon activation remain debated. The classical model predicts that PKA catalytic subunits dissociate from regulatory subunits in the presence of cAMP, whereas a second model proposes that catalytic subunits remain associated with regulatory subunits following physiological activation. Here we report that different PKA subtypes, as defined by the regulatory subunit, exhibit distinct subcellular localization at rest in CA1 neurons of cultured hippocampal slices. Nevertheless, when all tested PKA subtypes are activated by norepinephrine, presumably via the β -adrenergic receptor, catalytic subunits translocate to dendritic spines but regulatory subunits remain unmoved. These differential spatial dynamics between the subunits indicate that at least a significant fraction of PKA dissociates. Furthermore, PKA-dependent regulation of synaptic plasticity and transmission can be supported only by wildtype, dissociable PKA, but not by inseparable PKA. These results indicate that endogenous PKA regulatory and catalytic subunits dissociate to achieve PKA function in neurons.

eLife assessment

In this **important** study, Xiong and colleagues studied PKA regulation in synaptic plasticity. They provide **convincing** evidence that dissociation of PKA catalytic subunits is essential for the proper function of the kinase. Experiments using a PKA regulatory-catalytic subunit fusion establish that dissociative activation is required for both structural long-term potentiation and basal priming of AMPA receptors.

Introduction

Cyclic adenosine monophosphate (cAMP)-dependent kinase, or protein kinase A (PKA), regulates diverse critical functions in nearly all mammalian cells, including neurons. PKA is a tetrameric protein consisting of two regulatory subunits (PKA-Rs) and two catalytic subunits (PKA-Cs) (Francis

and Corbin, 1994 [↗](#); Johnson et al., 2001 [↗](#)). In the inactive state, each PKA-R binds to and inhibits a PKA-C. Binding of cAMP to PKA-R activates PKA-C. However, there are different proposals on the molecular events that follow activation.

For decades, PKA-C is thought to dissociate from PKA-R upon cAMP binding (Beavo et al., 1974 [↗](#); Francis and Corbin, 1994 [↗](#); Gold, 2019 [↗](#); Johnson et al., 2001 [↗](#); Reimann et al., 1971 [↗](#)). Freed PKA-C molecules then move to phosphorylate their substrates. However, a small number of studies (reviewed in Gold, 2019 [↗](#)), including two notable recent publications (Smith et al., 2017 [↗](#), 2013 [↗](#)), propose an alternative model, in which physiological concentrations of cAMP can activate PKA-C but do not result in its dissociation from PKA-R. Testing these two models will not only elucidate the biophysical mechanism of PKA activation, but also have distinct implications in how PKA may achieve its specificity, which is thought to rely on spatial compartmentalization (Wong and Scott, 2004 [↗](#)).

We have previously found that the majority of type II β PKA, as defined by PKA-R, is anchored to microtubules in the dendritic shaft of hippocampal CA1 pyramidal neurons where PKA-RII β is bound to the abundant microtubule associated protein MAP-2 (Zhong et al., 2009 [↗](#)). Upon activation of the β -adrenergic receptor with norepinephrine, a fraction of PKA-C dissociated from PKA-RII β (Tillo et al., 2017 [↗](#)). The freed PKA-C redistributed into dendritic spines, whereas PKA-RII β remained anchored at the dendritic shaft (Tillo et al., 2017 [↗](#); Xiong et al., 2021 [↗](#)). These results are consistent with the classical PKA activation model. However, recent studies suggest that PKA-Rs other than PKA-RII β may be the more abundant isoforms in CA1 neurons (Church et al., 2021 [↗](#); Ilouz et al., 2017 [↗](#); Weisenhaus et al., 2010 [↗](#)). It remains untested whether these PKA isoforms dissociate upon physiologically relevant stimulations in neurons.

Here, we examined whether PKA-C dissociates from the major PKA-R isoforms, PKA-RI β and PKA-RII α , in CA1 neurons. The rescue of function following knockdown of PKA-C was compared between wildtype dissociable PKA and an inseparable PKA variant in which PKA-C is covalently linked to PKA-R. The result supports the classical model of PKA activation via dissociation.

Results

Recent studies have suggested that PKA-RII α and PKA-RI β may be the prevalent isoforms in hippocampal CA1 neurons. Therefore, we co-expressed C-terminally EGFP tagged PKA-C (PKA-C-EGFP) (Tillo et al., 2017 [↗](#); Zhong et al., 2009 [↗](#)) and a cytosolic marker (mCherry) with either PKA-RII α or PKA-RI β in CA1 neurons of organotypic hippocampal slice cultures (**Figs. 1A** [↗](#) and **1B** [↗](#), upper left panels). At rest, PKA-C-EGFP exhibited a distribution that was dependent on the co-expressed PKA-R: when co-expressed with PKA-RII α , PKA-C was enriched in dendritic shafts; when co-expressed with PKA-RI β , PKA-C was more evenly distributed (quantified using the spine enrichment indexes, or SEI, see Methods) (**Fig. 1C** [↗](#)). This distribution was independent of the expression level (rest conditions in **Figure S1** [↗](#)) and largely resembled that of the corresponding PKA-R (**Figs. 1A-1C** [↗](#)).

Notably, upon application of norepinephrine (10 μ M), PKA-C of both subtypes translocated to dendritic spines, but the subcellular localization of PKA-Rs remained unchanged (**Figs. 1D** [↗](#) & **1E** [↗](#)). This differential re-distribution between PKA-C and PKA-R was more prominent following activation with forskolin (25 μ M) and IBMX (50 μ M). The translocation of PKA-C was independent of the expression level and the effect remained when extrapolating to the zero-overexpression level using a linear fit (**Figure S1** [↗](#)). These results can only be explained if at least a fraction of PKA-C dissociated from both PKA-RII α and PKA-RI β when a physiological stimulant was used. Together with our earlier results regarding PKA-C/PKA-RII β , we conclude that PKA-C dissociates from PKA-R with physiologically relevant stimuli.

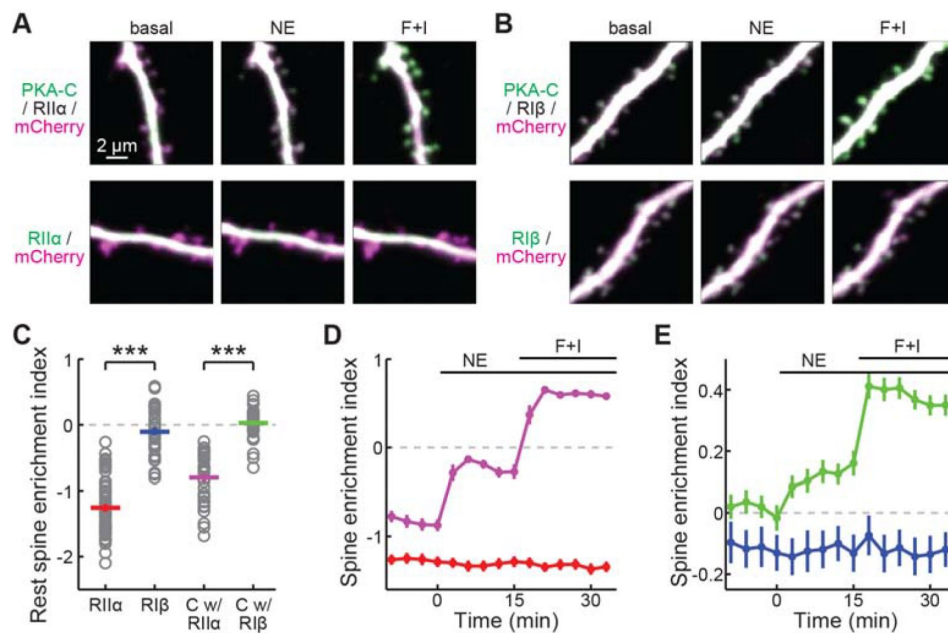


Figure 1.

PKA-C but not PKA-R redistributes to spines upon activation. (A, B) Representative two-photon images of PKA-C-EGFP co-expressed with PKA-RII α or PKA-RI β at rest, or in the presence of norepinephrine (NE) or forskolin and IBMX (F + I). mCherry (magenta) was co-expressed to reveal the neuronal morphology. (C-E) Quantification and comparison of the spine enrichment index at the resting state (C) and upon activation (D & E). As in panel C from left to right, n (spines/neurons) = 53/11, 34/7, 33/6, and 36/7.

Next, we asked whether PKA regulation of neuronal function is dependent on the dissociation of PKA-C from PKA-R. A key experiment supporting the non-dissociating PKA activation model was that PKA regulation of cell growth could be sustained by a construct in which PKA-C was fused to PKA-RII α in one polypeptide chain via a flexible linker (named R-C; **Fig. 2A**) (Smith et al., 2017). We therefore asked whether this R-C construct could support PKA regulation of neuronal function. When R-C-EGFP was expressed in CA1 neurons, this construct exhibited a distribution highly similar to that of RII α (**Figs. 2B** and **2C**). The tendency of this construct to translocate to the spine was largely diminished compared to PKA-C-EGFP co-expressed with wildtype, unlinked PKA-RII α (**Fig. 2D**), indicating that the catalytic subunit in R-C was indeed inseparable from the regulatory subunit.

To evaluate the function of R-C, a previously established shRNA construct was used to selectively knock down PKA-C α in CA1 neurons in cultured hippocampal slices (Tillo et al., 2017). Given that PKA activation is required for the late phase of long-term potentiation (L-LTP) (Abel et al., 1997), we examined the structural LTP of individual dendritic spines of CA1 neurons elicited by focal two-photon glutamate uncaging (**Fig. 3**) (Matsuzaki et al., 2004). The shRNA knockdown of PKA-C resulted in attenuated LTP at 90 min after induction (**Figs. 3A–3C**). This attenuation was not observed when a control shRNA against LacZ was expressed (**Fig. 3C**). The attenuated structural LTP was rescued by co-expression of shRNA-resistant wild-type PKA-C-EGFP together with PKA-RII α (**Figs. 3A–3C**). However, the R-C construct in which PKA-C was also resistant to shRNA knockdown but could not leave PKA-RII α failed to rescue the phenotype.

PKA activity has also been shown to regulate synaptic transmission. We therefore examined evoked AMPA and NMDA receptor (AMPA and NMDAR, respectively) currents in paired, transfected and adjacent untransfected CA1 neurons in cultured hippocampal slices. As shown previously (Tillo et al., 2017; Xiong et al., 2021), neurons expressing the shRNA construct against PKA-C exhibited significantly lower AMPAR currents (**Fig. 4A**), but not NMDAR currents (**Fig. 4B**). As a result, the AMPAR/NMDAR current ratio was also reduced (**Fig. 4C**). The deficits were rescued by co-expression of shRNA-resistant, wild-type dissociable PKA-C-EGFP and PKA-RII α (**Fig. 4**). However, the inseparable R-C construct failed to rescue the phenotype. Taken together, the R-C construct did not support normal PKA-dependent synaptic function.

Discussion

Our results indicate that at least a fraction of PKA-C molecules dissociate from all tested PKA-R isoforms, including I β , II α and previously tested II β , when activated by physiological stimuli. Given that PKA activity increases by two orders of magnitude when dissociated from PKA-R (Miyamoto et al., 1969), even a small fraction of PKA-C dissociation will result in a marked increase of PKA kinase activity. These results corroborate the observations in intact non-neuronal cells that used FRET imaging and biochemical measurements, respectively (Walker-Gray et al., 2017; Zaccolo et al., 2000). Furthermore, PKA-C that is covalently linked to PKA-RII α cannot functionally replace wildtype PKA for normal neuronal transmission or plasticity. Although this R-C construct has been shown to functionally replace endogenous PKA in terms of supporting the growth of a heterologous cell line (Smith et al., 2017), it cannot support all necessary PKA functions. Overall, we conclude that PKA-C dissociation from PKA-R is essential for PKA regulation of neuronal function. Additionally, PKA specificity is mediated by spatial compartmentalization (Wong and Scott, 2004). This is likely mediated by mechanism downstream of PKA dissociation, such as membrane tethering of freed PKA-C or its buffering by extra PKA-Rs (Gaffarogullari et al., 2011; Tillo et al., 2017; Walker-Gray et al., 2017; Zhang et al., 2015).

In addition, this study shows that PKA distribution in neuronal dendrites at the resting state is subtype dependent. PKA-RII α is enriched in dendritic shaft in a way similar to PKA-RII β , likely via its interaction with the abundant microtubule binding protein MAP2 (Tillo et al., 2017; Vallee et

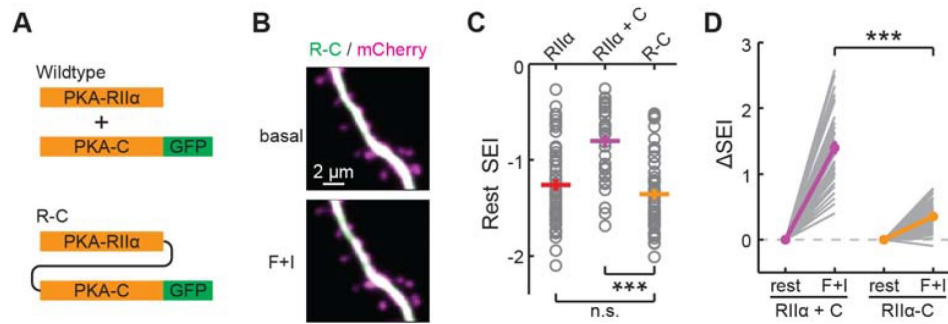


Figure 2.

Characterization of the inseparable R-C. (A) Schematic of wildtype PKA versus R-C. In both cases PKA-C was C-terminally tagged by mEGFP. (B–C) Representative images (B), quantifications of resting distribution (C), and the distribution change upon stimulation by forskolin and IBMX (D) of R-C compared to PKA-RIIα-EGFP and co-expressed PKA-C-EGFP/PKA-RIIα. RIIα and RIIα+C data are from **Fig. 1C**. n (spines/neurons) = 48/10.

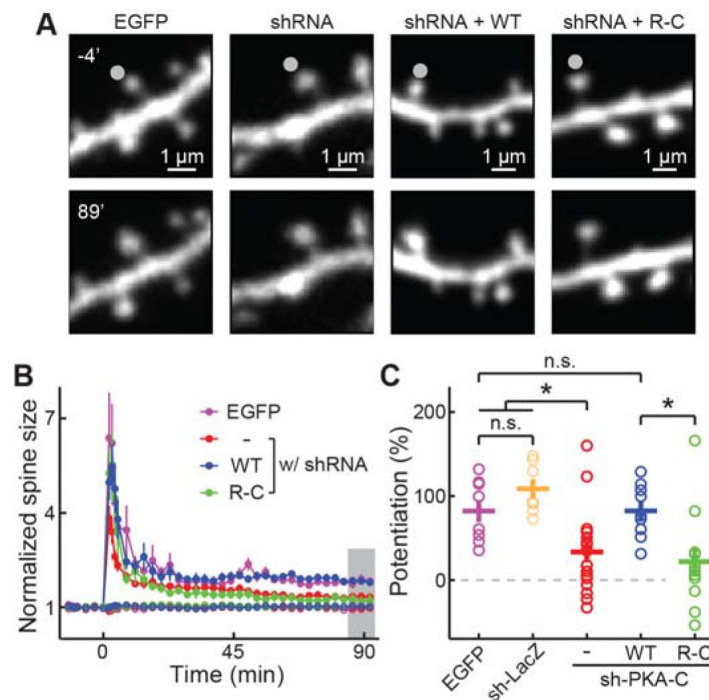


Figure 3.

PKA regulation of synaptic plasticity cannot be sustained by an inseparable PKA variant. (A–C) Representative image (A), time course (B), and the degree of potentiation (C) at the indicated timepoints in panel B of single-spine LTP experiments as triggered by focal glutamate uncaging at the marked spines (gray dot). In panel B, both stimulated spines (solid circles) and non-stimulated control spines (open circles) are shown. As in panel C from left to right, n (spines, each from a different neuron) = 8, 7, 17, 11, 9.

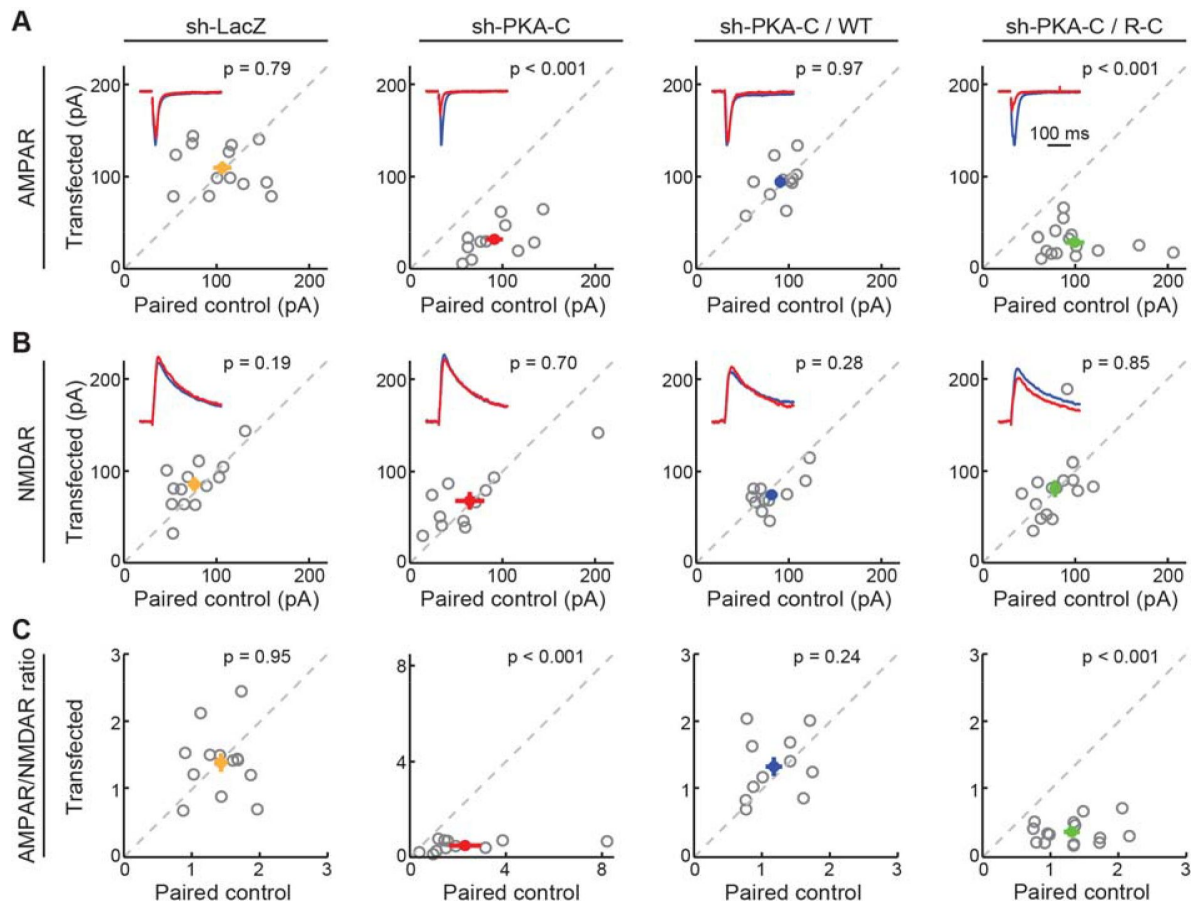


Figure 4.

AMPA receptor-mediated synaptic transmission requires wildtype dissociable PKA. (A–C) Representative traces (red) normalized to the paired control (blue) (insets) and scatter plots of paired AMPA (A) and NMDA (B) receptor currents and AMPA/NMDA receptor current ratios (C) from neighboring untransfected CA1 neurons and those transfected with shRNA against PKA-C and the indicated shRNA-resistant rescue constructs. Statistical p values were obtained using a sign test (MATLAB). From left to right, n (neuron pairs) = 13, 11, 11, and 15.

al., 1981 [\[1\]](#)). Note that this observation does not exclude the importance a small fraction of PKA being anchored to synaptic sites via other PKA binding proteins. In contrast, PKA-R1β is more evenly distributed between spine and dendrites. These observations establish a spatial organizational base for understanding subtype-specific PKA function in neurons.

This study also demonstrates that PKA is essential for long-term structural LTP of individual spines. PKA has been shown to facilitate the induction of LTP (i.e., metaplasticity) (Thomas et al., 1996 [\[2\]](#); Zhong et al., 2009 [\[3\]](#)), and is required for the maintenance of L-LTP (Abel et al., 1997 [\[4\]](#)), as assayed using electrophysiological recording of postsynaptic currents. However, it has been suggested that the structural and synaptic current changes may not be causally linked (Kopeck et al., 2006 [\[5\]](#)). Our results fill the gap to show that PKA is also required for maintaining the late phase of structural LTP.

Materials and Methods

Plasmid constructs

Constructs were made using standard mutagenesis and subcloning methods. All previously unpublished constructs and their sequences will be submitted to Addgene. In the R-C construct, mouse PKA-R1α and PKA-Cα were fused via a linker with residues WDPGSGSLEAGCKNFFPRFTSCGSLEGGSA AAA that were previously used (Smith et al., 2017 [\[6\]](#)).

Organotypic hippocampal slice cultures and transfections

Cultured rat hippocampal slices were prepared from P6 – P8 (typically P7) pups, as described previously (Stoppini et al., 1991 [\[7\]](#); Zhong et al., 2009 [\[3\]](#)). Animal experiments were performed in accordance with the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health, and were approved by the Institutional Animal Care and Use Committee (IACUC) of the Oregon Health & Science University (#IP00002274). cDNA constructs were transfected after 1.5–3 weeks *in vitro* via the biolistic gene transfer method using the Helios gene gun and 1.6 μm gold beads (Fig. 1 [\[8\]](#)) or, where long-term expression (~ 1 week) was required, with single-cell electroporation (Fig. 2 [\[9\]](#)) (Otmakhov and Lisman, 2012 [\[10\]](#)).

Two-photon imaging and two-photon glutamate uncaging

A custom built two-photon microscope based on an Olympus BW51WI microscope body was used. Laser beams from two different Ti:Sapphire lasers (Maitai, Newport) were aligned to allow for simultaneous two-photon excitation and photoactivation. Laser intensities were controlled by Pockels cells (Conoptics). Imaging and photoactivation were controlled by ScanImage (Vidrio Tech) (Pologruto et al., 2003 [\[11\]](#)). Slices were perfused during imaging in gassed artificial cerebral spinal fluid (ACSF) containing (mM) 127 NaCl, 25 NaHCO₃, 25 D-glucose, 2.5 KCl, 4 MgCl₂, 4 CaCl₂, and 1.25 NaH₂PO₄ with 0.5 μM tetrodotoxin (TTX). mEGFP fluorescence (green) was unmixed from that of the cytosolic marker (mCherry or DsRed Express) using a dichroic (Chroma 565DCXR) and band-pass filters (Chroma HQ510/70 for green and Semrock FF01-630/92 for red).

For single-spine structural LTP experiments, 2.25 mM MNI-caged-L-glutamate (Tocris) was added to ACSF containing 4 mM calcium, 0.05 mM magnesium, 1 μM TTX and 4 μM 2-chloroadenosine, as previously described (Harvey et al., 2008a [\[12\]](#)). To trigger structural plasticity, 30 pulses of 4-ms 16-mW (at back focal plane) 720-nm laser light were delivered to the spine head at 0.5 Hz.

Image analysis was performed using custom software written in MATLAB called *SI_View* (https://github.com/HZhongLab/SI_View [\[13\]](#)) (Ma et al., 2022 [\[14\]](#)). Using the software, regions of interest (ROIs) were manually drawn to isolate spines or their immediately adjacent dendritic shaft. Only the

spines well isolated from the dendrite laterally throughout the entire experiments were included. Spine enrichment index was calculated as:

$$SEI = \log_2[(F_{\text{green}}/F_{\text{red}})_{\text{spine}}/(F_{\text{green}}/F_{\text{red}})_{\text{shaft}}]$$

in which F is the average fluorescence intensity in an ROI.

The expression level was estimated by the maximal intensity from thick apical dendrite near the soma (typically 50–100 μm) similar to previously described (Harvey et al., 2008b [Tillo et al., 2017](#)). To minimize the influence of noise, a line of 9 pixels (pixel size $\sim 0.04 \mu\text{m}$) thick was manually drawn transecting the dendrite through the visually identified, brightest region. The line profile was further smoothened by 5 pixels along the line and baseline subtracted before determining the maximal value. To combine measurements from different hardware configuration (e.g., different microscopes), the data from each configuration and experiment were internally corrected for laser stimulation intensity and then normalized to the average of all data under the same condition.

Electrophysiology

Whole-cell voltage-clamp recordings were performed using a MultiClamp 700B amplifier (Molecular Devices). Electrophysiological signals were filtered at 2 kHz and digitized and acquired at 20 kHz using custom software written in MATLAB. Slices were perfused with artificial cerebrospinal fluid containing 4 mM Ca and 4 mM Mg. The internal solution contained (in mM) 132 Cs-gluconate, 10 HEPES, 10 Na-phosphocreatine, 4 MgCl_2 , 4 $\text{Na}_2\text{-ATP}$, 0.4 Na-GTP , 3 Na-ascorbate, 3 QX314, and 0.2 EGTA with an osmolarity of 295 mOsmol/kg. The junction potential was calculated to be -17 mV using a built-in function in the Clampfit software (Molecular Devices). Several less abundant anions (phosphocreatine, ATP, GTP and ascorbate) were omitted in the calculation due to lack of data in the program. The Cl reversal potential was -75 mV.

To reduce recurrent activities, cultured hippocampal slices were cut on both sides of CA1 and 4 μM 2-chloroadenosine (Sigma) was present in all recording experiments. 10 μM GABAzine (SR 95531, Tocris) was also included to suppress GABA currents. For electrical stimulation, a bipolar, θ -glass stimulating electrode (Warner Instruments) was positioned in the stratum radiatum 100–150 μm lateral to the recorded neuron. For all recordings, a transfected neuron and an untransfected neuron located within 50 μm of each other were sequentially recorded without repositioning the stimulation electrode. Measurements were carried out on averaged traces from approximately 20 trials under each condition. For AMPAR currents, the cells were held at -60 mV (before correcting for the junction potential) and the current was measured as the baseline-subtracted peak current within a window of 2–50 ms after electric stimulation. For NMDAR currents, the average currents at 140 to 160 ms after stimulation were used when the cells were held at +55 mV (before correcting for the junction potential).

Data analysis, presentation, and statistics

Quantification and statistical tests were performed using custom software written in MATLAB. Averaged data are presented as mean $\pm$ s.e.m., unless noted otherwise. p values were obtained from one-way ANOVA tests, unless noted otherwise. In all figures, *: $p \leq 0.05$ and is considered statistically significant after Bonferroni correction for multiple tests, **: $p \leq 0.01$, and ***: $p \leq 0.001$.

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Author contributions

H.Z. conceived the project. W.X. and H.Z. designed the experiments. W.X. and H.Z. performed the experiments and data analyses. M.Q. produced critical reagents used in the experiments. H.Z. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Figures and figure legends

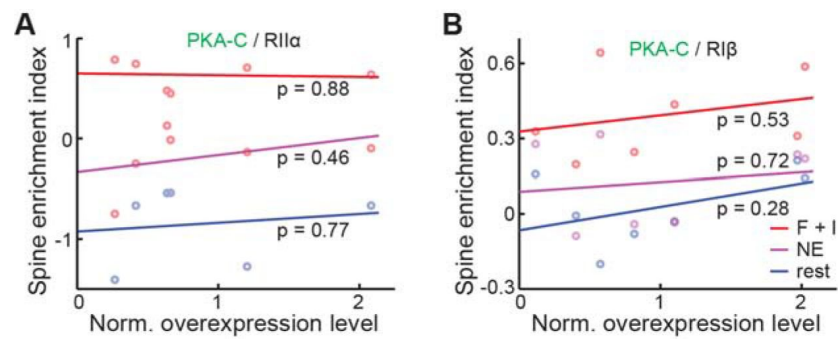


Figure S1.

Spine enrichment indexes and the movement of PKA-C into spines upon activation are not dependent on the protein expression level. (A–B) Averaged spine enrichment indexes within individual neurons of PKA-C co-expressed with RIIα (A) and RIβ (B), and their linear fit at rest or under the indicated stimulation.

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Reviewer #1 (Public Review):

Summary:

This is a short self-contained study with a straightforward and interesting message. The paper focuses on settling whether PKA activation requires dissociation of the catalytic and regulatory subunits. This debate has been ongoing for ~ 30 years, with renewed interest in the question following a publication in *Science*, 2017 (Smith et al.). Here, Xiong et al demonstrate that fusing the R and C subunits together (in the same way as Smith et al)

prevents the proper function of PKA in neurons. This provides further support for the dissociative activation model - it is imperative that researchers have clarity on this topic since it is so fundamental to building accurate models of localised cAMP signalling in all cell types. Furthermore, their experiments highlight that C subunit dissociation into spines is essential for structural LTP, which is an interesting finding in itself. They also show that preventing C subunit dissociation reduces basal AMPA receptor currents to the same extent as knocking down the C subunit. Overall, the paper will interest both cAMP researchers and scientists interested in fundamental mechanisms of synaptic regulation.

Strengths:

The experiments are technically challenging and well executed. Good use of control conditions e.g untransfected controls in Figure 4.

Weaknesses:

The novelty is lessened given the same team has shown dissociation of the C subunit into dendritic spines from RIIbeta subunits localised to dendritic shafts before (Tillo et al., 2017). Nevertheless, the experiments with RII-C fusion proteins are novel and an important addition.

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Reviewer #2 (Public Review):

Summary:

PKA is a major signaling protein that has been long studied and is vital for synaptic plasticity. Here, the authors examine the mechanism of PKA activity and specifically focus on addressing the question of PKA dissociation as a major mode of its activation in dendritic spines. This would potentially allow us to determine the precise mechanisms of PKA activation and address how it maintains spatial and temporal signaling specificity.

Strengths:

The results convincingly show that PKA activity is governed by the subcellular localization in dendrites and spines and is mediated via subunit dissociation. The authors make use of organotypic hippocampal slice cultures, where they use pharmacology, glutamate uncaging, and electrophysiological recordings.

Overall, the experiments and data presented are well executed. The experiments all show that at least in the case of synaptic activity, the distribution of PKA-C to dendritic spines is necessary and sufficient for PKA-mediated functional and structural plasticity.

The authors were able to persuasively support their claim that PKA subunit dissociation is necessary for its function and localization in dendritic spines. This conclusion is important to better understand the mechanisms of PKA activity and its role in synaptic plasticity.

Weaknesses:

While the experiments are indeed convincing and well executed, the data presented is similar to previously published work from the Zhong lab (Tillo et al., 2017, Zhong et al 2009). This reduces the novelty of the findings in terms of re-distribution of PKA subunits, which was already established. A few alternative approaches for addressing this question: targeting localization of endogenous PKA, addressing its synaptic distribution, or even impairing within intact neuronal circuits, would highly strengthen their findings. This would allow us

to further substantiate the synaptic localization and re-distribution mechanism of PKA as a critical regulator of synaptic structure, function, and plasticity.

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Reviewer #3 (Public Review):

Summary:

Xiong et al. investigated the debated mechanism of PKA activation using hippocampal CA1 neurons under pharmacological and synaptic stimulations. Examining the two PKA major isoforms in these neurons, they found that a portion of PKA-C dissociates from PKA-R and translocates into dendritic spines following norepinephrine bath application. Additionally, their use of a non-dissociable form of PKC demonstrates its essential role in structural long-term potentiation (LTP) induced by two-photon glutamate uncaging, as well as in maintaining normal synaptic transmission, as verified by electrophysiology. This study presents a valuable finding on the activation-dependent re-distribution of PKA catalytic subunits in CA1 neurons, a process vital for synaptic functionality. The robust evidence provided by the authors makes this work particularly relevant for biologists seeking to understand PKA activation and its downstream effects essential for synaptic plasticity.

Strengths:

The study is methodologically robust, particularly in the application of two-photon imaging and electrophysiology. The experiments are well-designed with effective controls and a comprehensive analysis. The credibility of the data is further enhanced by the research team's previous works in related experiments. The conclusions of this paper are mostly well supported by data. The research fills a significant gap in our understanding of PKA activation mechanisms in synaptic functioning, presenting valuable insights backed by empirical evidence.

Weaknesses:

The physiological relevance of the findings regarding PKA dissociation is somewhat weakened by the use of norepinephrine (10 μ M) in bath applications, which might not accurately reflect physiological conditions. Furthermore, the study does not address the impact of glutamate uncaging, a well-characterized physiologically relevant stimulation, on the redistribution of PKA catalytic subunits, leaving some questions unanswered.

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